Family Income and the Gap in Educational Attainment: Evidence from Higher Education

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Motivation

- A college degree is still widely seen as the path to a good job, a high salary, and generally a better life
- But some students face barriers even **attending** college, or struggle to **finish** once they start
- Family income is a commonly cited factor:
  - Access to tutors and better-funded secondary schools
  - Greater ability to pay for college
  - Low-income families face credit constraints (Plug and Vijverberg 2005)
- **What is the independent effect of family income on the probability a child attends and graduates college?**

Literature Review

- **Becker and Tomes (1979); Taubman (1989):** Theory says altruistic parents invest in a child's education until its marginal return equals the return on financial assets $\Rightarrow$ richer families can invest more
- **Haveman and Wolfe (1995):** Over half of the early literature finds a positive, significant effect of parental income, though small relative to family structure, moves, and siblings
- **Plug and Vijverberg (2005):** Using adoptees as a natural experiment, family income has a significant *causal* effect on years of education and college graduation
- **Haveman and Wilson (2005):** PSID cohort born 1966–1970; top-quartile children are almost $4\times$ as likely to graduate and $3\times$ as likely to attend college as bottom-quartile children

This Paper's Contribution

An update and extension of Haveman and Wilson (2005):

	Haveman & Wilson	This Paper
Cohort born	1966–1970	1993–1997
Data through	1995	2023
Wealth measured, ages	14–18	16–20
Race/ethnicity	Black vs. all others	Hispanic, Black, Other (White non-Hisp. baseline)
Regional analysis	No	Yes

- Also tests whether the income effect differs by race, ethnicity, and region via interactions

Data

- **Panel Study of Income Dynamics (PSID)**, University of Michigan
- Panel collapsed to a cross section: one observation per child summarizing their entire childhood
- PSID became biennial in 1997, so age-varying variables are matched to the closest wave
- Tuition data from the National Center for Education Statistics
- All dollar values in 2014 dollars (CPI-U)

- **Sample:**
  - 1,262 individuals unweighted; 1,219 with PSID sample weights
  - 994 observations used in the regressions

Variables

Outcomes (measured at age 25):

- `attended_college` = 1 if $\geq$ 13 years of education
- `grad_college` = 1 if $\geq$ 16 years of education

Literature: following explanatory variables have persistent and statistically significant relation with educational attainment

Key regressor: `log_inc`, log of average family income-to-needs, ages 2–15

- Income-to-needs = real family income / Census poverty threshold for that family size
- Captures long-run income rather than a few points in time

Variables

Explanatory variables cont:

- Wealth in 2013 (log positive wealth, negative wealth)
- Race/ethnicity dummies, sex dummies
- Parental education (graduated HS, attended college, missing)
- Total location moves and average siblings (ages 2–15)
- In-state public 4-year tuition per FTE, 2014 ($1,000s)
- Census region dummies (Midwest is baseline in the model)

Summary Statistics (Selected)

Variable	Weighted Mean	Unweighted Mean
Attend college	0.546	0.480
College graduate	0.365	0.269
Years of education	13.94	13.63
Family income-needs	3.656	2.879
Real family wealth, 2013	$253,210	$149,734
Hispanic	0.237	0.146
Black non-Hispanic	0.155	0.448
Female	0.444	0.515
At least one parent attended college	0.516	0.439
Number of location moves	1.795	2.367
Average number of siblings	1.432	1.451
Tuition and fees per FTE ($1,000s)	9.124	8.995

Raw Gaps by Income and Wealth Quartile

Family Income

Q	Attend	Grad	Yrs
1st	34.73	18.26	12.90
2nd	43.63	21.33	13.43
3rd	44.10	20.82	13.48
4th	77.92	65.01	15.10

Family Wealth

Q	Attend	Grad	Yrs
1st	30.94	12.59	12.89
2nd	32.77	10.92	12.88
3rd	52.33	25.97	13.82
4th	73.91	57.31	14.94

- Top income quartile graduates at $\approx 3.5\times$ the rate of the bottom; attends at $\approx 2.25\times$
- Top wealth quartile graduates at $4.55\times$ the rate of the bottom
- Versus Haveman and Wilson, the top–bottom **graduation** gap grew by about 10 percentage points; the **attendance** gap shrank

Econometric Model

Logit regression, following Haveman and Wilson (2005):

$$\begin{aligned} y_i = \alpha &+ \beta\,\text{log_inc}_i + \gamma\,\text{log_pos_wealth}_i + \zeta\,\text{neg_wealth}_i \\ &+ \delta\,\text{hispanic}_i + \theta\,\text{black}_i + \eta\,\text{other}_i + \omega\,\text{female}_i \\ &+ \mu\,\text{parent_hs}_i + \lambda\,\text{parent_col}_i + \tau\,\text{parent_edu_miss}_i \\ &+ \phi\,\text{moves}_i + \rho\,\text{tuition}_i + \xi\,\text{south}_i + \psi\,\text{northeast}_i + \nu\,\text{west}_i \\ &+ \kappa\,\text{interactions}_i + \varepsilon_i \end{aligned}$$

- $y_i \in \{\text{grad_college},\ \text{attended_college}\}$
- Interactions: `log_inc` $\times$ race/ethnicity and `log_inc` $\times$ region
- Baselines: White non-Hispanic and Midwest
- Omits H&W's single-parent, neighborhood dropout, and wealth-missing variables $\Rightarrow$ possible omitted variable bias

Baseline Logit Results

	Graduated College	Attended College
Log of family income needs	1.034^{***}	0.711^{***}
	(0.248)	(0.215)
Log of positive wealth 2013	0.086^{**}	0.057^{*}
	(0.038)	(0.030)
Hispanic	1.090^{***}	1.025^{***}
	(0.357)	(0.336)
Black non-Hispanic	0.442	0.465
	(0.324)	(0.290)
Female	0.533^{**}	0.933^{***}
	(0.229)	(0.224)
At least one parent attended college	1.159^{***}	1.043^{***}
	(0.330)	(0.302)
Total moves	-0.157^{**}	-0.175^{***}
	(0.077)	(0.065)
Tuition per FTE 2014	0.015	0.038
	(0.055)	(0.060)

Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Siblings and other controls not shown (insignificant).

Does the Income Effect Differ by Race and Ethnicity?

	Grad	Attend
Log inc	1.562^{***}	0.812^{***}
Hispanic	2.498^{***}	1.328^{***}
Black non-Hisp.	1.410^{**}	0.604
Other non-Hisp.	-0.584	-0.037
Log inc × Hisp.	-1.560^{***}	-0.514
Log inc × Black	-0.864	-0.166
Log inc × Other	1.677^{*}	1.114

- The family income effect **does not increase** the probability Hispanic students graduate college
- Black non-Hispanic becomes significant and positive, but no evidence the income effect differs
- For attendance, no evidence the income effect differs by race or ethnicity

Separating ethnicity from race (vs. H&W's single Black dummy) uncovers variation that more restrictive models miss.

Does the Income Effect Differ by Region?

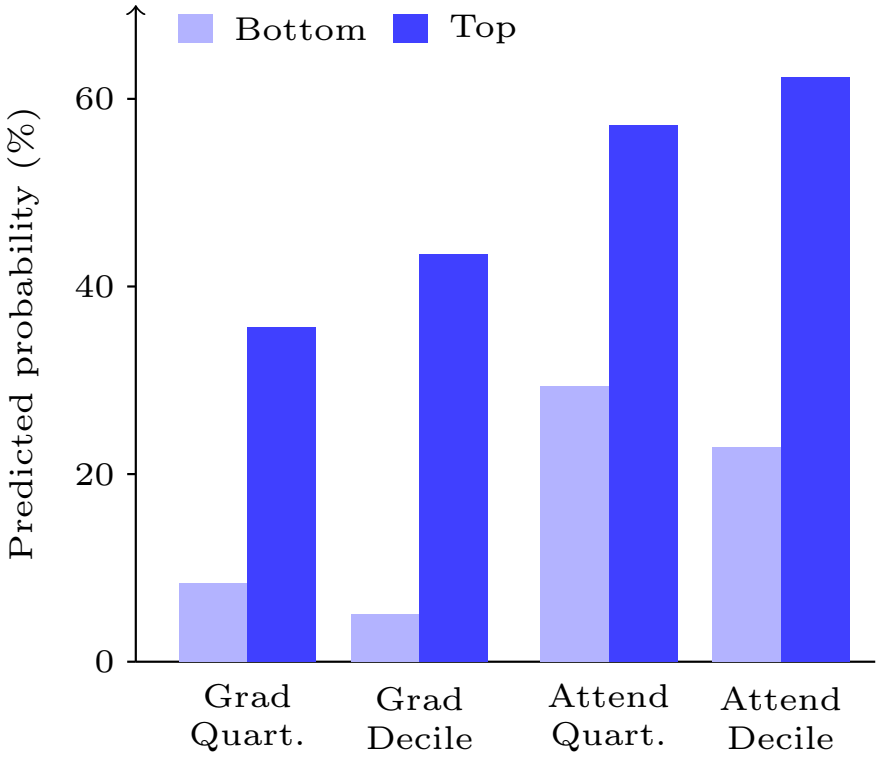
- The Northeast leads all regions in income, tuition, and attainment: 67% attended and 47% graduated college
- Full model adds region dummies and `log_inc` × region interactions

	Grad	Attend
Log inc	1.566^{***}	0.766^{**}
South	−0.145	−0.597
Northeast	0.438	0.749
West	−0.361	−0.652
Log inc × Northeast	−0.194	−0.502
Log inc × West	0.103	0.332
Log inc × South	-1.58	-0.653

- All region terms and interactions are insignificant: no evidence that outcomes or the income effect differ by region after controls

Predicted Educational Attainment by Family Income

	Bottom Q	Top Q
Graduate (%)	8.35	35.71
Attend (%)	29.42	57.18
Years of educ.	12.47	13.89
	Bottom D	**Top D**
Graduate (%)	5.05	43.43
Attend (%)	22.91	62.36
Years of educ.	12.06	14.16



- Top quartile: $> 4\times$ as likely to graduate, almost $2\times$ as likely to attend, > 1 more year of education
- Top decile: $> 8\times$ as likely to graduate, almost $3\times$ as likely to attend

Robustness: Probit Models

- Re-estimate every logit model as a probit
- Main results hold; only some controls shift in significance:
  - Total moves becomes marginally significant in baseline graduation model ($p = 0.070$)
  - Log positive wealth becomes significant in baseline attendance model
  - Log inc $\times$ Other non-Hispanic and positive wealth becomes significant in graduation models

Probit Predicted Values (%)

	Bottom	Top
Quartiles		
Graduate	6.94	35.57
Attend	27.43	57.53
Deciles		
Graduate	3.67	43.25
Attend	20.61	62.93

Top-group predictions are almost exactly reproduced; bottom-group predictions fall by 1–2 points, so every gap **widens**.

Discussion

- Family income remains significantly and positively associated with college attendance and graduation, net of family, demographic, and tuition controls
- Since Haveman and Wilson (2005):
  - The **graduation** gap has widened
  - The **attendance** gap has shrunk
  - Gaps are largest at the extremes of the income distribution
- The income effect on graduation is smaller for Hispanic students; no evidence it varies by region

Discussion

- **Implications:** Education carries a large wage premium and many nonmarket benefits (Wolfe and Haveman 2002)
  - Higher child cognitive ability
  - Better health outcomes
  - Lower rates of teen pregnancy
  - Better health
  - Higher life expectancy
  - Better ability to deal with stress
  - Increased economic independence
- Taken with casual evidence, children from poor families face mobility challenges from the start
- "The nation's colleges and universities appear to be an integral part of the process whereby family economic status is passed along from generation to generation."